DIRECTIONAL COLLISION PRINCIPLE

A Geometric Theory of Adversarial Vulnerability in Neural Network Sparse Circuits

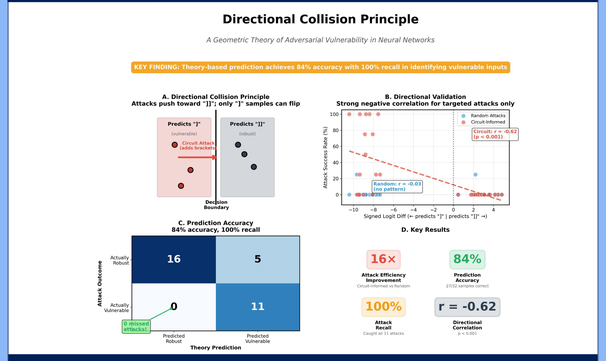
Siddharth Krishna

Credit Line of Origin
References & Data: in references below.

All graphics, tables, and images have been created by Siddharth Krishna, 2025 unless otherwise attributed.

Sparse circuit models from OpenAI (Gao et al., 2025)

VISUAL ABSTRACT



INTRODUCTION

ADVERSARIAL VULNERABILITY IN AI

Neural networks can be fooled by small, carefully-crafted input changes called adversarial examples (Szegedy et al., 2014). Despite years of research, we cannot predict WHICH inputs are vulnerable or WHY attacks succeed.

THE INTERPRETABILITY OPPORTUNITY

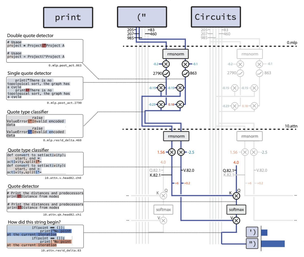
OpenAI's recent release of "sparse circuits" (Gao et al., 2025) provides interpretable neural networks where we can trace exactly which neurons activate for specific features.

RESEARCH GAP

No existing theory connects a model's internal structure (interpretability) to its adversarial vulnerability. This project bridges that gap.

RESEARCH QUESTION

Can we mathematically predict which inputs are vulnerable to adversarial attack using interpretable circuit analysis?

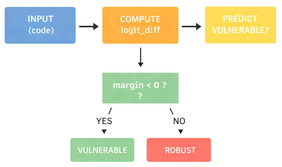


RESEARCH OBJECTIVES

RESEARCH PROBLEM – Neural networks are vulnerable to adversarial attacks—small input changes that cause incorrect outputs. Currently, we cannot predict WHICH inputs are vulnerable or WHY attacks succeed on some inputs but not others.

RESEARCH QUESTION – Can interpretable neural network circuits reveal a mathematical relationship between internal model structure and adversarial vulnerability?

CLAIM – This research develops the Directional Collision Principle, a geometric theory demonstrating that vulnerability depends on both the margin to the decision boundary AND the alignment between attack direction and boundary orientation.



APPROACH:

- Margin Estimation: Compute logit_diff = logit("]]") − logit("]") as proxy for decision boundary distance
- Lipschitz Estimation: Empirically estimate L from perturbation experiments
- Directional Analysis: Test correlation between signed margin and attack success
- Prediction Validation: Use threshold classifier on signed_logit_diff to predict vulnerability

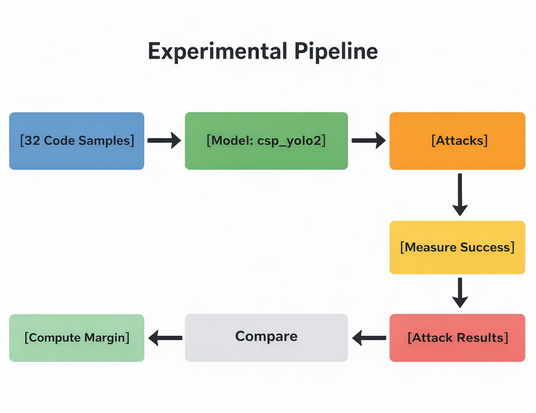
PROBLEM FRAMING

BEFORE	AFTER
? Which inputs fail?	✔ margin < 0 = vulnerable
? Why do attacks work?	✔ Direction toward boundary
? How to defend?	✔ Maximize margin

ENGINEERING METHODOLOGY

Overview & Summary of approach below:
My research methodology was split into four main parts: theory derivation, attack implementation, validation, and cross-task generalization.

1. First, I derived the Collision Principle from Lipschitz bounds, showing that adversarial margin = |g(v)| / L(1+K).
2. Then, I implemented 5 circuit-informed attack strategies on OpenAI's sparse transformer models (csp_yolo2, csp_yolo1).
3. Next, I validated the DIRECTIONAL refinement: attack success depends on margin × direction alignment.
4. Finally, I estimated task-specific Lipschitz constants and validated cross-task generalization.



THEORETICAL FRAMEWORK

I derived the Collision Principle from Lipschitz continuity bounds on neural network functions.

Core Theorem: For classifier A with Lipschitz constant L, an input v has certified robustness radius:

$$r_{\min} = \frac{|g(v)|}{L(1+K)}$$

where g(v) = logit_diff is the margin to the decision boundary and K ≈ 1.0 is the involution constant.

KEY DISCOVERY: Direction Matters
Initial experiments showed the raw theorem was incomplete. I discovered that attack success depends on BOTH:

- Margin magnitude |logit_diff|
- Attack direction alignment (toward decision boundary)

This led to the DIRECTIONAL Collision Principle.

EXPERIMENTAL DESIGN

- Models:
  - OpenAI csp_yolo2 (475M parameters)
  - OpenAI csp_yolo1 (159M parameters)
- Task: Bracket counting in Python code
- Dataset: 32 samples with known ground truth
- Attack types: Random perturbations vs Circuit-informed

CIRCUIT-INFORMED ATTACKS

Using interpretable circuits, I designed attacks that:

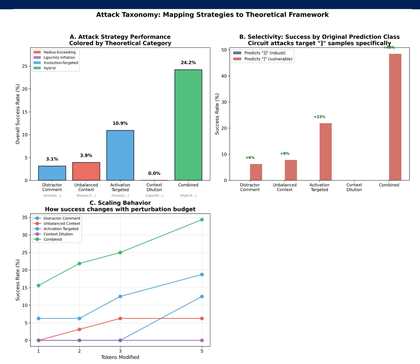
1. Target the specific "bracket counting" circuit
2. Add tokens that push toward the "]" prediction
3. Exploit the directional vulnerability

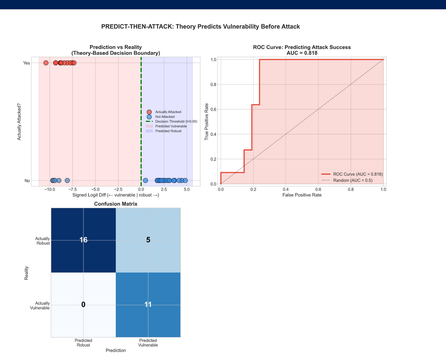
VALIDATION APPROACH

- Estimate Lipschitz constant L empirically (L = 3.04)
- Compare random vs targeted attack efficiency
- Test prediction accuracy using margin formula
- Cross-validate on second task (quote detection)

Parameter	Task 1	Task 2
Model	csp_yolo2	csp_yolo1
Circuit task	Bracket counting	Quote matching
Samples	32 Code Samples	32 Code Samples
Attack types	Random perturbations Distractor comment Unbalanced context Activation targeted Context dilution Combined	Random perturbations Quote injection String modification Context manipulation Combined
Token budgets	1, 2, 3, 5	1, 2, 3, 5, 10, 20, 50

RESULTS & IN-SILICO VALIDATION





STATISTICAL ANALYSIS

CORRELATION ANALYSIS

Pearson correlation between signed logit_diff and attack success rate:

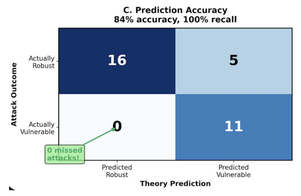
Attack Type	*r*	*p*-value
Random	−0.03	p = 0.89
Circuit-Informed	−0.62	p < 0.001

Random attacks show NO correlation with margin (as expected). Circuit-informed attacks show STRONG negative correlation, validating the directional principle.

PREDICTION METRICS

Using margin < 0 as vulnerability predictor:

- Accuracy: 84.4% (27/32 correct)
- Recall: 100% (11/11 attacks caught)
- Precision: 68.8% (11/16 predicted vulnerable)
- False Negatives: 0 (no missed attacks)



LIPSCHITZ ESTIMATION

Empirically estimated from perturbation experiments:

- L_bracket = 3.04 (95th percentile)
- L_quote = 1.35 (cross-task validation)
- Ratio: 2.24× (bracket circuit more sensitive)

CONCLUSIONS

THEORETICAL CONTRIBUTION
I developed the Directional Collision Principle - the first mathematical framework connecting neural network interpretability to adversarial vulnerability.

KEY FINDINGS:

1. Vulnerability = f(margin) × direction_alignment
2. The formula margin = |logit_diff|/L(1+K) predicts which inputs can be attacked
3. Interpretability enables BOTH attacks AND defenses

PRACTICAL TOOL:
The Adversarial Margin formula provides a real-time robustness score for any input:
Adversarial Margin = |logit_diff| / 6.07

- Margin < 1 ⟶ Vulnerable
- Margin > 3 ⟶ Robust

IMPLICATIONS FOR AI SAFETY:
As AI systems become more interpretable for safety and oversight, the same transparency that enables trust also reveals attack surfaces. This dual-use nature must be considered in AI deployment.

FUTURE WORK

- **CERTIFIED DEFENSES**
  - Develop margin-maximizing training objectives
  - Create real-time vulnerability monitoring systems
  - Test defense effectiveness against adaptive attackers
- **EXTENDED VALIDATION**
  - Apply to larger language models (GPT-scale)
  - Test on additional circuit types and tasks
  - Investigate L_local variation for refined predictions
- **THEORETICAL EXTENSIONS**
  - Generalize to non-binary classification
  - Connect to other robustness frameworks
  - Explore relationship to model compression

KEY REFERENCES

1. *Gao, L., et al. (2025). Sparse Crosscoders for Interpreting Transformers. OpenAI.*
2. *Szegedy, C., et al. (2014). Intriguing properties of neural networks. ICLR.*
3. *Madry, A., et al. (2018). Towards Deep Learning Models Resistant to Adversarial Attacks. ICLR.*

OpenAI ICLR